

M 237/62

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Total No. of printed pages = 04

Monsoon, 2024

B. Tech 1st Semester Examination

Basic Electrical Engineering

Course Code: BEL23101T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

Section-I

Answer all the questions from this section

Marks: 10x1=10

Q.1 (a) Which of the following relations is not correct?

(i) $P = \frac{V}{R^2}$ (ii) $P = VI$ (iii) $I = \sqrt{\frac{P}{R}}$ (iv) $V = \sqrt{PR}$

(b) The rms value of a sine wave is 100 A. Its peak value is
(i) 70.7 A (ii) 141 A (iii) 150 A (iv) 282.2 A

(c) Which of the following essential features is possessed by an indicating instrument?
(i) Deflecting torque (ii) Controlling torque (iii) Damping torque (iv) All of the above

(d) A silicon PN junction in forward conduction has a voltage drop closer to
(i) 1.7 V (ii) 2.1 V (iii) 1.7 V (iv) 0.7 V

(e) What is the primary function of an MCB (Miniature Circuit Breaker)?
(i) To regulate voltage
(ii) To protect the circuit from overloading and short circuits
(iii) To increase current in the circuit
(iv) To convert AC to DC

(f) A fuse material should possess the following properties
(i) Low melting point and low specific resistance
(ii) High melting point and low specific resistance
(iii) Low melting point and high specific resistance
(iv) High melting point and high specific resistance

(g) Two sinusoidal emfs are given as
 $e_1 = A \sin\left(\omega t + \frac{\pi}{4}\right)$ and $e_2 = B \sin\left(\omega t - \frac{\pi}{6}\right)$
The phase difference between the two quantities in degrees is
(i) 75° (ii) 105° (iii) 60° (iv) 15°

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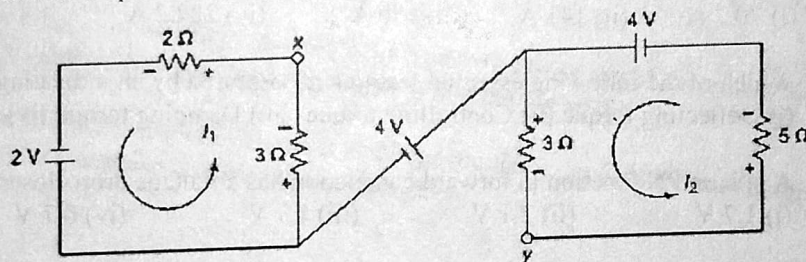
- (h) The maximum rectification efficiency of a half-wave rectifier is
 (i) 40.6% (ii) 81.2% (iii) 88% (iv) 100%
- (i) A practical current source is usually represented by
 (i) A resistance in series with an ideal current source
 (ii) A resistance in parallel with an ideal current source
 (iii) A resistance in parallel with an ideal voltage source
 (iv) None of the above
- (j) The most efficient form of damping employed in electric instruments is
 (i) Air friction damping
 (ii) Fluid friction damping
 (iii) Eddy current damping
 (iv) None of the above

Section-2

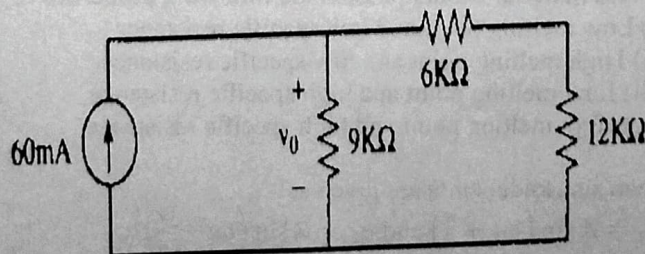
Answer any one question from this section

Marks: 10

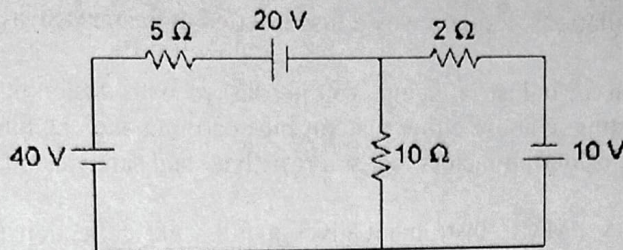
- Q.2 (a) (i) Solve the given DC circuit using Kirchhoff's Voltage Law (KVL) to find the potential drop between the points x and y.



- (ii) Solve the circuit and find the voltage V_0 , the power supplied by the current source and the power absorbed by each element in the following circuit



- (b) Apply Thevenin's theorem to calculate the value of current flowing through the $2\ \Omega$ resistor



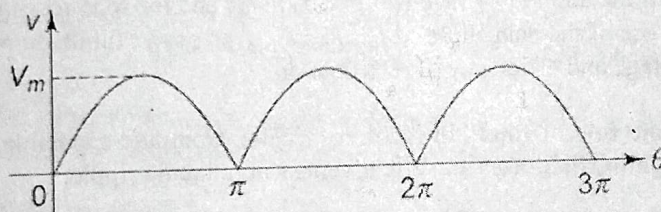
Section-3

Answer any one question from this section

Marks: 10

- Q.3 (a) (i) Examine the Faraday's Law of Electromagnetic Induction to determine how it can be applied to calculate the instantaneous voltage generated in a rectangular coil with N turns and area A when rotating at an angular velocity ω in a uniform magnetic field B .

- (ii) Make use of the following waveform to find the average value of the waveform



- (b) (i) Two sinusoidal waveforms are given as $i_1 = 5 \cos(\omega t + 30^\circ) \text{ A}$ and $i_2 = -10 \cos(\omega t + 60^\circ) \text{ A}$. Examine the summation of currents in polar and rectangular form.

- (ii) A single phase AC rms voltage V of 50 Hz frequency is applied to a series circuit consisting of resistance R and inductance L . Assuming voltage as a reference, determine the expressions for impedance, current and power factor. Draw an impedance triangle and phasor diagram.

Section-4

Answer any one question from this section

Marks: 10

- Q.4 (a) (i) A moving iron instrument is used to measure the current in an electrical circuit. Examine the working of the attraction-type principle of the instrument when connected to an AC supply.

- (ii) List the advantages and disadvantages of using MCB over fuse in domestic installations. Explain why a fuse inserted in the phase wire is never in the neutral wire?
- (b) (i) In an industrial setup, you are tasked with designing an earthing system for the building. Choose either plate or pipe earthing and explain how you would implement it, considering factors like soil resistivity and safety standards.
- (ii) A PMMC instrument gives a full-scale deflection (fsd) of 100 mA when the potential difference across its terminals is 100 mV. Make use of the data and find the shunt resistance for an fsd corresponding to 100 A, also the resistance for an fsd with 100 V.

Section-5

Marks: 10

Answer any one question from this section

- Q.5 (a) (i) Construct a full-wave bridge rectifier circuit and explain its working. Identify its advantages over a half-wave rectifier.
- (ii) A crystal diode having internal resistance $R_f = 20\Omega$ is used for half-wave rectification. If the applied voltage is $v = 50 \sin \omega t$ and the load resistance $R_L = 800\Omega$. Make use of the data and find I_m , I_{dc} , I_{rms} , ac power input, dc power output, dc output voltage and efficiency of rectification.
- (b) (i) Examine the role of filters in rectifier circuits. Compare capacitor and inductor filters, highlighting their uses and effectiveness in reducing ripple.
- (ii) Explain how a Zener diode is used as a voltage regulator. Include a circuit diagram to show how it maintains a stable output voltage.